

From: Brady, Donal (EGLE)
Sent: 9/29/2023 11:53:59 AM
To: "Patrick Miller" <pmiller@cadillac-mi.net>; "Cindy Tomaszewski" <ctomaszewski@Cadillac-MI.net>
Cc: "Jeff Dietlin" <utilities@cadillac-mi.net>
Subject: Cadillac WWTP - NPDES Permit MI0020257 - July 10, 2023 Inspection
Attachments: Cadillac WWTP Jul 2023 WRD CSI Inspection Form 071023.pdf, 2023 Cadillac WWTP CSI Final Sampling Report 082223.pdf

Hi Patrick and Cindy,

This email is the formal transmittal for the inspection we completed earlier this year. The results of the sampling EGLE completed along with the City are also attached.

On July 10, 2023, staff of the Michigan Department of Environment, Great Lakes, and Energy (EGLE), Water Resources Division (WRD), conducted an evaluation of Cadillac wastewater treatment plant (WWTP), located at 1121 Plett Road in Cadillac, Wexford County, to evaluate the facility's compliance with Part 31, Water Resources Protection, of the Natural Resources and Environmental Protection Act (1994 PA 451, as amended) and National Pollutant Discharge Elimination System (NPDES) Permit No. MI0020257. The facility (with an average day design flow of 3.2 MGD) discharges municipal wastewater, through Outfall 001, to the Clam River.

Patrick Miller and Cindy Tomaszewski from the City of Cadillac and Don Brady from EGLE, WRD participated in the inspection, which included an interview, records review, laboratory review, and a site walkthrough.

The WWTP and WWTP laboratory are well maintained and other items evaluated during the inspection were acceptable and in compliance with the permit. I appreciate the work you and the City are doing to make changes and improvements to the WWTP. Thanks for all the great work and for your time during the inspection. The inspection form includes a list of minor things that I/we should follow-up on. I will spend more time on that list in the coming weeks, after the end of EGLE's fiscal year on September 30. If you have any questions or want to talk about this email, please let me know.



Donal Brady, P.E.
Environmental Engineer
Water Resources Division / Cadillac District Office
Michigan Department of Environmental, Great Lakes, and Energy
120 West Chapin Street, Cadillac, MI 49601
231-429-5686 cell | BradyD6@michigan.gov
[Follow Us | michigan.gov/EGLE](https://michigan.gov/EGLE)

ATTACHMENT NAME:

Cadillac WWTP Jul 2023 WRD CSI Inspection Form 071023.pdf

ATTACHMENT TYPE:

Adobe Portable Document Format (PDF) compound image

Cadillac WWTP : 15637

NPDES - Facility Site Review

Inspector: Donal Brady

Start Date: 07/10/2023

Inspection Information

1. Permit Number: MI0020257

2. Date: 07/10/2023

3. Time: 1:00 PM

4. Inspection Participants Present

Name	Title	Affiliation	Phone Number
Patrick Miller	Water Resources Supervisor	Cadillac	231-306-8084
Cindy Tomaszewski	Laboratory Supervisor	Cadillac	231-779-8089
Don Brady	Environmental Engineer	EGLE	231-429-5686

5. Opening Conference Notes:

We discussed operational changes since last inspection

- influent screw pump was repaired
- influent grinder pump fixed
- the smallest primary clarifier was taken out of service (the retention time in the primary clarifiers was too high, approx. 6 hours)
- the plant started wasting into this unused primary clarifier vs. co-settling and this improved plant performance

We also discussed future work planned

- piston sludge pump replacement
- new headworks building
- SCADA upgrade

6. Select the current facility classification(s). See Help instructions if there is no facility classification on record.

Class A, treatment facilities serving or designed to serve a population in excess of 50,000 persons.

The WWTP serves a population between 10,000 and 15,000 people but was upgraded to Class A due to plant complexity and the discharge to a trout stream.

7. Designated Certified Operator

Certification Type	Operator Name	Number	Exp. Date	Reported to DEQ?
A, B, C, D	Jeff Dietlin	10435	07/15/2026	X
B, C, D	Patrick Miller	19313	01/15/2025	X

8. If the facility is a mobile home park, do any residents own their own lot? ☐ Yes ☐ No ☒ NA

9. If applicable, population served by treatment plant (systems that serve the public only) 11,377

Per 2022 NPDES renewal application

10. Has the service area of wastewater plant changed since last application submittal? (public system only) ☐ Yes ☒ No ☐ NA

Cadillac WWTP : 15637

NPDES - Facility Site Review

Inspector: Donal Brady

Start Date: 07/10/2023

Inspection Information

11. Identify chemicals used in treatment system

Chemical Name	Use/Purpose	Concentration	Location Feed Point	Authorized	None used
Ferric chloride	Phosphorus removal	38.5%	Aeration / activated sludge basins	X	
Sodium hypochlorite	Periodic treatment of filamentous	90%	Various	X	

The WWTP may evaluate moving the FeCl feed to the downstream end of the basins.

12. If the facility is a POTW, does it have an approved Industrial Pretreatment Program (IPP)? ☒ Yes ☐ No ☐ NA

13. If the POTW accepts septage, landfill leachate or other high strength waste, via truck or sewer, and does not have an IPP, how does it determine it can adequately treat this waste?

The WWTP has an IPP and is currently in the process of reevaluating the headworks loading.

14. Number of wastewater personnel recommended in the O&M Manual (public systems) 1

Answer to #14 NA, the manual discusses shifts not # of personnel.

15. If the facility accepts septage, is it fully treated in the POTW? ☐ Yes ☐ No ☐ NA

The WWTP does not accept septage. The WWTP has 19 industrial user facilities (14 different companies) per the IPP. The facility also accepts floor wash water from High Point Auto and receives water from vacuum truck waste (from sewer cleaning).

16. Does the facility have compliance issues that may be related to the acceptance of septage? ☐ Yes ☐ No ☒ NA

17. Does the facility have any plans to change any of their operations as it relates to the acceptance of septage? ☐ Yes ☒ No ☐ NA

18. What is the facility's onsite solids storage capacity design in months? Approximately 1 year

19. What is the estimated remaining space currently in the storage structure? 9 months

20. Solids are disposed of through? (choose all that apply): Land application

21. If the facility normally land applies, is there a back-up disposal plan in case they are unable to land apply? The back up plan would be landfill disposal (currently grit is disposed in a landfill)

22. Have they had any operation or compliance issues that were related to biosolids storage issues? ☐ Yes ☒ No ☐ NA

Cadillac WWTP : 15637

NPDES - Facility Site Review

Inspector: Donal Brady

Start Date: 07/10/2023

Specific Processes Producing Wastewater (Industrial/Commercial Only) & Treatment Unit Processes

23. If an industrial/commercial facility, identify the types of wastewater generated.

This is a municipal WWTP

24. If an industrial/commercial facility, are these wastewater sources identified in the current permit application?

☐ Yes ☐ No ☒ NA

25. List the specific industrial/commercial processes that produce the raw wastewater flows:

Process Name

Municipal WWTP

General Observations/Questions

26. No. of wastewater treatment personnel

7

27. Does the number of wastewater treatment personnel appear to be adequate?

☒ Yes ☐ No

28. Number of hours each shift and/or day that the waste treatment system is staffed.

Shift #	Sun	Mon	Tues	Wed	Thur	Fri	Sat
Shift 1	Few hours	7 AM to 3:30 PM	7 AM to 3:30 PM	7 AM to 3:30 PM	7 AM to 3:30 PM	7 AM to 3:30 PM	Few hours
Shift 2							
Shift 3							

29. How are alarms/upsets monitored and responded to during off hours and on weekends?

telephone auto-dialer

An auto-dialer call the City offices during business hours and central dispatch during off hours, WWTP personnel are subsequently notified by the City or central dispatch

30. List all treatment unit processes, including solids handling

activated sludge, clarification, disinfection, equalization basin, aeration, attached growth, other (note other treatment below in comments), Sludge storage, Sludge stablization

Attached growth = RBCs used for tertiary treatment, sludge stabilization = anaerobic digestion, disinfection = UV, other = cloth filtration, downstream of RBCs, for tertiary treatment

Cadillac WWTP : 15637

NPDES - Facility Site Review

Inspector: Donal Brady

Start Date: 07/10/2023

General Observations/Questions

31. Are all treatment units in service? If No, explain in comments below, including why, how long out of service, and when expected to be back in service.

☐ Yes ☒ No ☐ NA

One primary clarifier taken off-line as discussed earlier, formerly out of service grinder and screw pump both fixed

32. Any SSOs of the collection system or bypasses at the WWTP since the last inspection?

☐ Yes ☒ No ☐ NA

33. If yes, what is the date of the SSO/bypass and the date notification was made to EGLE?

Date of bypass

Date notification made to DEQ

34. If the facility is a POTW, are effluent flows at or above 80% of the design capacity?

☐ Yes ☒ No ☐ NA

35. Are hydraulic and/or organic overloads experienced? If Yes, explain in comments.

☐ Yes ☒ No ☐ NA

The WWTP sees occasional spikes in flow and/or loading (above the long term averages) from industry and from seasonal population increases.

36. Has there been an increase in flow since your last permit was issued?

☐ Yes ☒ No

37. If an industrial or privately owned facility, is the current effluent flow rate greater than that authorized by the current permit?

☐ Yes ☐ No ☒ NA

38. Is a permit modification needed?

☐ Yes ☒ No

39. Identify the influent and effluent flow monitoring points.

Monitoring Point	Location	Acceptable
Influent	Headworks building after grinder	X
Effluent	After tertiary filters	X

40. Influent flow is measured before all return lines? (public system only) If No, where is it measured?

☒ Yes ☐ No ☐ NA

41. The effluent flow measurement represents all wastewater being discharged (e.g. after all return lines)?

☒ Yes ☐ No

The effluent flow also includes cloth filter backwash.

42. Select type of effluent flow metering

weir

43. When was it last calibrated?

08/15/2022

The meter was calibrated after the 2022 date, Patrick needs to check the actual date with another operator.

Cadillac WWTP : 15637

NPDES - Facility Site Review

Inspector: Donal Brady

Start Date: 07/10/2023

General Observations/Questions Cont.

44. Has the facility received any citizen complaints or experienced any spills since the last inspection? ☐ Yes ☒ No ☐ NA

45. Any unusual conditions noted in treatment system effluent, receiving water, irrigation fields, infiltration basins, or tile fields? If Yes, describe in comments section. ☐ Yes ☒ No ☐ NA

46. Any evidence of bypasses or chemical spills, surcharges and/or overflow in the wastewater collection or treatment system? If Yes, describe in comments section. ☐ Yes ☒ No ☐ NA

47. Any accumulation of solids, grease, foam or floating material in wetwells, sumps, clarifiers, land application areas, or other treatment units? ☐ Yes ☒ No ☐ NA

48. Any excessive discoloration, puddles, oil or grease on ground or floor near equipment or tanks? ☐ Yes ☒ No ☐ NA

49. Identify in the table below all significant chemical storage areas.

Chemical Name	Location	Concentration	Volume
Ferric chloride	Room/building next to the UV	38.5%	Two tanks each approx. 8,200 gallons

50. Does the Facility conduct observations of their chemical use and storage areas to determine if leaks or discharges have occurred? ☒ Yes ☐ No ☐ NA

51. Does secondary containment for outdoor storage of chemicals appear to have sufficient capacity, be compatible with the material stored in it, and be capable of containing the material should a spill occur? ☐ Yes ☐ No ☒ NA

52. Does indoor storage of chemicals appear to have sufficient curbing, containment, or is situated such that a spill would not reach public sewer systems or surface / groundwaters? ☒ Yes ☐ No ☐ NA

53. Does the facility have a Pollution Incident Prevention Plan (PIPP) or Integrated Contingency Plan (ICP)? ☒ Yes ☐ No

WRD will send the WWTP information on Part 5 notifications.

54. Any problems with wastewater treatment chemical feed system? If Yes, describe in comments section. ☐ Yes ☒ No ☐ NA

55. Has the facility reported chemical spills that have occurred? ☐ Yes ☐ No ☒ NA

56. Any jury rigged or temporary systems to correct operational problems? If Yes, describe in comments ☐ Yes ☒ No ☐ NA

57. Any obnoxious odors noted? If Yes, describe from where in comments. ☐ Yes ☒ No ☐ NA

58. Were any unidentified/unmarked confined spaces observed? If Yes, describe where in comments. ☐ Yes ☒ No ☐ NA

Cadillac WWTP : 15637

NPDES - Facility Site Review

Inspector: Donal Brady

Start Date: 07/10/2023

General Observations/Questions Cont.

59. Is lighting adequate for safe operation and observation?	☒ Yes ☐ No
60. Are aisles, ladders, stairs and other access points kept clear and in good condition?	☒ Yes ☐ No ☐ NA
61. Does general housekeeping appear to be adequate?	☒ Yes ☐ No
62. Does the treatment system security/fencing appear to be adequate?	☒ Yes ☐ No ☐ NA
63. Were photographs taken?	☐ Yes ☒ No
64. Have all deficiencies previously cited in inspections, enforcement actions, correspondence and verbal communications been corrected?	☒ Yes ☐ No ☐ NA
65. Is a permit modification in order due to a change in name, ownership, or operations?	☐ Yes ☒ No ☐ NA

Closing Information

66. Completed By:	Don Brady
67. Closing Conference Notes: WWTP will check on flow meter calibration. WRD will send information on PIPP notifications. WRD will send asset management examples and WWTP will provide additional info. WRD will also send WET test lab alternatives. WRD and WWTP will follow up regarding DMR QA explanations. WRD and WWTP will check on DMR discrepancies. WRD will check on WTA approval needed for hypochlorite. WRD will return to WWTP to observe Hach P analytical method. WWTP will work to implement a re-weight procedure for TSS analysis. WRD and WWTP will follow up on DMR/QA 42 documentation.	

Cadillac WWTP : 15637

NPDES - Self-Monitoring Records

Inspector: Donal Brady

Start Date: 07/10/2023

General Information

1. Facility Designated Name:	Cadillac WWTP
2. Permit Number:	MI0020257
3. Date:	07/10/2023
4. Time:	1:00 PM

Record Information

5. Are all records and information resulting from the monitoring activities required by this permit maintained for at least three years? Examples include DMR's, MORs, lab data, chain of custody forms, calibration & maintenance records and recordings from continuous monitoring instrumentation.	☒ Yes ☐ No ☐ NA
---	---

Use the table(s) in the following section to spot check at least three sets of data points (including geometric means, loadings, averages, etc.).

Analytical Tables

6. Use the table below to spot check sets of data points (including geometric means, loadings, averages, etc.).						
Date	Sampling Point:	Type of Wastewater	Parameter	Reported Result (from DMR)	Site Record (from bench sheet or lab)	Accurate?
05/26/2023	After tertiary	Municipal	P	0.198 mg/l	0.221 mg/l	
05/26/2023	After tertiary	Municipal	TSS	6 mg/l	6 mg/l	X
05/26/2023	After tertiary	Municipal	NH4	0.30 mg/l	0.101 mg/l	
05/26/2023	After tertiary	Municipal	CBOD	12.3 mg/l	12.3 mg/l	X
05/26/2023	Outfall	Municipal	pH	8.15	8.15	X
05/26/2023	Outfall	Municipal	DO	8.65	8.65	X
05/26/2023	Outfall	Municipal	Fecal coliform	1	<1	X

7. Are analytical results/bench sheets consistent with the data reported on the DMRs? Explain any discrepancies in the comments.	☐ Yes ☒ No ☐ NA
--	---

Errors appear to be data entry mistakes, WWRD will ask WWTP to confirm

Record Information Cont.

8. Does the facility have authorization for elimination of, or reduced monitoring? If yes, identify in the Comment Section which parameters have been reduced or eliminated along with their "new" monitoring frequency?	☐ Yes ☒ No
--	---

Cadillac WWTP : 15637

NPDES - Self-Monitoring Records

Inspector: Donal Brady

Start Date: 07/10/2023

Record Information Cont.

9. Monitoring Records are adequate and include:

Parameter	Check if included
Dates, times, frequency and location of sampling collection	X
Name of individual(s) performing sampling	X
Results of analyses	X
Dates of analyses	X
Name of person(s) performing analyses	X
Analytical technique or method used	X
The date of and name of person responsible for equipment calibration	X

10. Are laboratory bench sheets and other records understandable and clearly filled out? ☒ Yes ☐ No ☐ NA

11. Does the inspection contact/operator have a current copy of the permit? ☒ Yes ☐ No ☐ NA

12. Can the permittee produce a copy of their WTA approvals along with documentation that the factors associated with the approval have not changed (e.g. dosage rates, flow volumes, bench test calculations, etc.)? ☐ Yes ☐ No ☒ NA

FeCl WTA approved per Part 41, WRD will check on hypochlorite

Closing Information

13. Closing Conference Notes:
See facility site review form

14. Completed By: Don Brady

Cadillac WWTP : 15637

NPDES - Sampling Procedures

Inspector: Donal Brady

Start Date: 07/10/2023

General Information

1. Facility Designated Name: Cadillac WWTP
2. Permit Number: MI0020257
3. Date: 07/10/2023
4. Time: 1:00 PM

5. Sampling Parameters (if answer is not available, use comments section):

Parameter	Sampling Location	Sampling Equipment (auto-sampler, etc)	Sample Type (grab, composite, etc)	Sampling Location Representative?	Sampling Container Material and Quality	Preservation Method	Holding Time
NH3-N	After tertiary	Autosampler	Composite	X	HDPE	Cool	28 days
TSS	After tertiary	Autosampler	Composite	X	HDPE	Cool	7 days
(C) BOD5	After tertiary	Autosampler	Composite	X	HDPE	Cool	48 hours
Total Phosphorus	After tertiary	Autosampler	Comopsite	X	HDPE	H2SO4, cool	28 days
Fecal Coliform	Outfall		Grab	X	Polystyrene	Cool	8 hours
Hg	Outfall		Grab	X	Lab supplied	Cool	72 hours
pH	Outfall	Meter	Grab	X			15 minutes
DO	Outfall	Meter	Grab	X			15 minutes
Metals	After tertiary	Autosampler	Composite	X	Lab supplied	NO3, cool	6 months
WET	After tertiary	Autosampler	Composite	X	Lab supplied	Cool	
Bis-2-ethylhexylphthalate	After tertiary	Autosampler	Composite	X	Lab supplied	Cool	7 days
PFOS/PFOA	After tertiary		Grab	X	Lab supplied	Cool	14 days

WRD to confirm WET test holding time at next inspection.

Inspection Information

6. How are staff trained to collect samples? One-on-one w/ experienced staff, On-the-job training
7. Are sample collection procedures reviewed periodically? ☒ Yes ☐ No
8. Are sampling procedure changes made when appropriate? ☒ Yes ☐ No
9. Are 24-hour composite samples flow-proportional? If No, explain how it is representative of the discharge in the comments. ☒ Yes ☐ No ☐ NA
10. Has an alternate sample type been approved?

Parameter	Sample Type
NA	
11. Does the facility have an approved alternate monitoring location? ☐ Yes ☒ No

Composite Sampler Information

12. Monitoring Point Location: After tertiary

Cadillac WWTP : 15637

NPDES - Sampling Procedures

Inspector: Donal Brady

Start Date: 07/10/2023

Composite Sampler Information

13. Review and/or inspect the composite sampler(s).

Question	Yes	No	NA
Is the temperature in the composite sampler 0-6°C?	X		
Does the sample line maintain a minimum velocity of 2ft/s?	X		
Is a line purging cycle used between samples?	X		
Any obstructions in the sample line?		X	
Does the length or number of bends in the sample line interfere with obtaining a representative sample?		X	

WWTP obtained NIST thermometers

Inspection Information Cont.

14. Is the laboratory refrigerator, or any equipment used to store samples, kept at a temperature between 0-6°C? If not, explain in the comments. ☒ Yes ☐ No

15. Does the facility conduct sampling in accordance with the NPDES permit language or 40 CFR Part 136? ☒ Yes ☐ No

16. Is monitoring and analyses performed more often than required by the permit? ☐ Yes ☒ No ☐ NA

17. If Yes, are the results included in the facility's monthly reports? ☐ Yes ☐ No ☒ NA

Closing Information

19. Closing Conference Notes:
See facility site review form

20. Completed By: Don Brady

Cadillac WWTP : 15637
 NPDES - Analytical Methodology
 Inspector: Donal Brady
 Start Date: 07/10/2023

General Information

1. Date:	07/10/2023
2. Time:	1:00
3. Facility Designated Name:	Cadillac WWTP
4. Permit Number:	MI0020257

Laboratories

5. Identify all lab(s), including the in-house and contract labs, that perform the monitoring required by the NPDES Permit and list the parameters analyzed by that lab in the table below.

Name of Lab	Parameter(s)
Pace	Hg, Bis 2 ethylhexyl phthalate, biosolids - conventional parameters, PFAS
Eurofins	PFAS
Fibertec	PFAS
ERM	WET
HML	Biosolids - pathogens

6. Evaluate the chain of custody record required between the permittee and a contract laboratory for the following elements.

Facility name, Sample date, Sample location, Sampler name, Number of samples, Parameters to be analyzed, Preservative used, Intake temperature, Signatures, Dates and times

7. Does the contract lab have a QA/QC Program? ☒ Yes ☐ No ☐ NA

8. Does the permittee periodically review the QA/QC program associated with the analyses performed by the contract lab to ensure it is adequate? ☒ Yes ☐ No ☐ NA

Cadillac WWTP : 15637
 NPDES - Analytical Methodology
 Inspector: Donal Brady
 Start Date: 07/10/2023

Analytical Methods Used

9. Analytical Methods Table

Parameter	Method Used	Approved / Acceptable Method	Comments
CBOD	SM5210B	X	
TSS	SM2540D	X	
Ammonia	SM4500-NH3D	X	See below
Phosphorus	SM4500PE	X	See below
Fecal coliform	Colilert18	X	
PFOS/PFOA	EPA537.1(modified)	X	
Cu, Ni, Se	EPA200.8	X	
Bis 2-ethylhexyl phthalate	EPA625.1	X	
Hg	EPA1613E/EPA245.1	X	
pH	SM4500H+B	X	
DO	SM4500OG	X	

The WWTP is in the process of transitioning to Hach methods for phosphorus and ammonia, the WWTP is completing side-by-side analysis with the current P method and the Hach P method, the WWTP is also planning for the Hach NH4 method and investigating distillation equipment. The WWTP is still training staff and working to implement the TSS re-weigh.

10. Select one or more parameters to evaluate the protocols and/or techniques used during the analysis to determine if it meets the requirements of the test method.

Parameter	Testing protocols acceptable?
WRD will return to the WWTP next year to review the Hach P method	

11. Approval has been obtained for alternative test methods? ☐ Yes ☐ No ☒ NA
12. Facility has reference manuals for the test methods that they are using? ☒ Yes ☐ No ☐ NA
13. Facility has written procedures or manuals for instruments and equipment use. ☒ Yes ☐ No ☐ NA
14. Are laboratory calibration records maintained? ☒ Yes ☐ No ☐ NA

Instrumentation

15. List the analytical instruments the facility uses in their self-monitoring activities. *Confirm acceptability of calibration method using manufacturer specifications or test method requirements

Instrument	Calibration Method*	Frequency	Date last serviced
LBOD probe (luminous) for CBOD	Operator cal curve per method	Daily	Within last year
Balance	Third party contractor	Annually	Within last year
ICP mass spectrometer (metals)	Operator cal curve per method	Daily	Within last 6 months
Spectrophotometer (P and NH4) DR3900	Third party contractor	Annually	Within last year
pH/DO	Operator cal curve per method	Weekly	Within last year

16. Do laboratory instruments and equipment appear to be in good condition? ☒ Yes ☐ No ☐ NA

Cadillac WWTP : 15637
 NPDES - Analytical Methodology
 Inspector: Donal Brady
 Start Date: 07/10/2023

Instrumentation

17. Are records kept with dates of laboratory equipment maintenance/repair/replacement?	☒ Yes ☐ No ☐ NA																				
18. Proper volumetric glassware is used?	☒ Yes ☐ No ☐ NA																				
19. Glassware is properly cleaned and sterilized, if necessary?	☒ Yes ☐ No ☐ NA																				
20. Select the type of analytical balance used. If the balance used it "Other", describe it in the comment section.	Electronic																				
21. Is the balance clean and free of loose powders, stains, and corrosion?	☒ Yes ☐ No ☐ NA																				
22. If the balance is not equipped with automatic vibration compensation, is it mounted on a solid, vibration free surface, isolated from excessive heat, cold, or drafts?	☒ Yes ☐ No ☐ NA																				
23. Equipment temperature log maintained for ovens, incubators, etc.	☒ Yes ☐ No ☐ NA																				
24. Equipment temperatures. Indicate whether each is acceptable by checking Yes, No, or NA.																					
<table border="1"> <thead> <tr> <th>Equipment and allowable temperature</th> <th>Actual temperature</th> <th>Yes</th> <th>No</th> <th>NA</th> </tr> </thead> <tbody> <tr> <td>BOD incubator at 20°C? (±1°)</td> <td>19.7</td> <td>X</td> <td></td> <td></td> </tr> <tr> <td>Fecal coliform incubator at 44.5°C? (±0.2°)</td> <td>44.5</td> <td>X</td> <td></td> <td></td> </tr> <tr> <td>Drying oven for SS at 103-105°C? (±1°)</td> <td>104</td> <td>X</td> <td></td> <td></td> </tr> </tbody> </table>	Equipment and allowable temperature	Actual temperature	Yes	No	NA	BOD incubator at 20°C? (±1°)	19.7	X			Fecal coliform incubator at 44.5°C? (±0.2°)	44.5	X			Drying oven for SS at 103-105°C? (±1°)	104	X			
Equipment and allowable temperature	Actual temperature	Yes	No	NA																	
BOD incubator at 20°C? (±1°)	19.7	X																			
Fecal coliform incubator at 44.5°C? (±0.2°)	44.5	X																			
Drying oven for SS at 103-105°C? (±1°)	104	X																			
25. Has the dessicant used with the suspended solids analysis reached its moisture saturation point?	☐ Yes ☒ No ☐ NA																				
26. Are temperatures verified with a National Institute of Standards and Technology (NIST) thermometer or NIST verified thermometer?	☒ Yes ☐ No ☐ NA																				
The WWTP obtained digital NIST verified thermometers																					
Quality Assurance/Quality Control (QA/QC)																					
27. Are QA/QC written procedures in place for all required monitoring?	☒ Yes ☐ No ☐ NA																				
28. Are QA/QC results _____ on control charts?	recorded																				
Control charts and tables maintained for CBOD, P, and NH4																					
29. If data is plotted on control charts, does the data stay within upper and lower control limits?	☒ Yes ☐ No ☐ NA																				

Cadillac WWTP : 15637
 NPDES - Analytical Methodology
 Inspector: Donal Brady
 Start Date: 07/10/2023

Quality Assurance/Quality Control (QA/QC)

30. If QA/QC data are outside of control limits, what was done to identify the cause?

Replace reagents,
 Check equipment,
 Create a new
 phosphorus curve,
 Clean glassware, Check
 laboratory water,
 Review testing
 protocols, Review data
 for accuracy

31. Were the QA/QC issues adequately addressed?

☒ Yes ☐ No ☐ NA

32. Using the same parameters previously reviewed under the Analytical Methods section, what QA/QC methods are used? If not applicable, put NA in the first parameter box and delete all other pre-populated parameters.

Parameter	Duplicates Frequency	Spikes Frequency	Blank/GGA Frequency	Slope Check	Reference Samples Frequency	Inter-lab Splits Frequency
CBOD (D,GGA,R,I)	1x/week		1x/day		1x/year	Periodically
TSS (D,R,I)	1x/week		1x/week		1x/year	Periodically
Ammonia as N (D,S,Slope,R,I)	1x/week	1x/week	1x/week	1x/day	1x/year	Periodically
Total phosphorus (D,S,R,I)	1x/week	1x/week	1x/week	1x/week and cal curve 1x/6 weeks	1x/year	Periodically
pH (D,Slope,R)	1x/week			1x/week (cal curve)	1x/year	
Fecal coliform (D,B)	1x/week	1x/week (+check)	1x/week		1x/year	

33a. If an EPA major or selected minor, has the facility participated in the EPA DMR/QA program?

☒ Yes ☐ No ☐ NA

33b. If yes, characterize their result from the list below. Use the Comments to explain results, if needed.

Acceptable, Not
 Acceptable -
 Inadequate Corrective
 Action

DMR/QA 42 Not acceptable for Cu, Ni, Se due to a bad dilution, analysis redone and results provided, P also not acceptable analysis redone and results provided, WWTP will provide explanation of P results

34. Are there chemicals in the laboratory with an expired shelf life? If the chemical is associated with a parameter being monitored for the NPDES permit, identify the name of the chemical and the expiration date of that chemical in the Comment Section.

☐ Yes ☒ No ☐ NA

Laboratory Data Handling and Personnel

35. Significant figures are established for each analysis and round-off rules are uniformly applied?

☒ Yes ☐ No ☐ NA

Cadillac WWTP : 15637
NPDES - Analytical Methodology

Inspector: Donal Brady
Start Date: 07/10/2023

Laboratory Data Handling and Personnel

36. Does staffing appear to be sufficient? If No, provide an explanation in the Comment Section. ☒ Yes ☐ No ☐ NA

37. Is there sufficient time allowed for analyses, QA/QC, laboratory maintenance, and recordkeeping? If No, provide an explanation in the Comment Section. ☒ Yes ☐ No ☐ NA

38. How are lab analysts trained?" On-the-job training

39. Any changes in laboratory personnel since the last inspection? If Yes, identify what the change was and how (if at all) it has affected laboratory operations in the Comment Section. ☐ Yes ☒ No ☐ NA

Closing Information

40. Closing Conference Notes:
See facility site review form.

41. Completed By: Don Brady

Cadillac WWTP : 15637

NPDES - Areas Evaluated

Inspector: Donal Brady

Start Date: 07/10/2023

Areas Evaluated	
1. Permit Review	Not Applicable
2. Compliance Schedule	Not Applicable
3. Facility Site Review	Satisfactory
4. Effluent/Receiving Waters	Satisfactory
5. Records/Reporting (Off-site review)	Satisfactory
6. Self-Monitoring Program (On-site review)	Satisfactory
7. Sample Protocol	Satisfactory
8. Laboratory	Satisfactory
9. Biosolids/Sludge Disposal (Optional)	Not Evaluated
10. Flow Measurement (Optional)	Not Evaluated
11. Operation & Maintenance (Optional)	Not Evaluated
12. Other	Not Applicable

ATTACHMENT NAME:

2023 Cadillac WWTP CSI Final Sampling Report 082223.pdf

ATTACHMENT TYPE:

Adobe Portable Document Format (PDF) compound image

MICHIGAN DEPARTMENT OF ENVIRONMENT, GREAT LAKES and ENERGY
WATER RESOURCES DIVISION
POINT SOURCE MONITORING

Report of a
Municipal Wastewater Survey
Conducted at
Cadillac WWTP
NPDES No. MI0020257
COUNTY: Wexford
Cadillac, MI 49601
Start Date: July 11, 2023

SURVEY SUMMARY

Wastewater monitoring was performed during one, twenty-four hour sampling period starting on July 11, 2023 at 9:30 AM.

The results of the physical, chemical and bacteriological analyses are presented in Tables 1, 2 & 3.

The results of MDEQ monitoring are compared to the results reported by the facility in Tables 1, 2 & 3.

A composite sample from monitoring point 001A was split with the facility to compare laboratory analyses. The results are shown in Table 4.

The results of this survey are compared to the results of the previous survey performed in May of 2018 in Table 5.

Samples were preserved according to Table 6. Letter codes for laboratory results are defined in Table 6.

SURVEY PROCEDURE

The flow and samples were obtained as follows:

Sample Location	Flow Measurement	Sampling Methods
001A - Composite samples collected from outfall weir	Facility Analysis Report	Automatic composite samples, Grab composite samples
001A - Grab samples collected from outfall bypass	---	Individual grab samples
Upstream samples collected from Smith Street Bridge on the downstream side	---	Individual grab samples

An automatic sampler composites samples at timed intervals.

A grab composite consists of a series of individual grabs composited into one sample.

An individual grab is a single instantaneous sample.

Samples were analyzed by the EGLE Laboratory located in Lansing, MI and Merit Laboratories, Inc. in East Lansing, MI.

Table 1 - Composite Samples

	EGLE		Cadillac WWTP	
Monitoring Point	001		001	
Survey Period Start Date/Time	7/11/23 10:00		7/11/2023 7:00	
Survey Period End Date/Time	7/12/23 9:45		7/12/2023 7:00	
Flow Rate (MGD)	1.92		1.92	
	Results		Results	
	mg/L	lbs/day	mg/L	lbs/day
Suspended Solids	4	64.05	4.4	70.46
CBOD-5	<3 ND	NA	3	48.04
BOD-5	<3 ND	NA	---	NA
COD	37	592.47	---	NA
TOC	12	192.15	---	NA
NO2	0.012	0.19	---	NA
NO3 + NO2	9.3	148.92	---	NA
NH3	0.03	0.48	<0.3	<4.80
Kjeldahl Nitrogen	1.2	19.22	---	NA
Total Phosphorus	0.4	6.41	0.3	4.80
Ortho Phosphate	0.23	3.68	---	NA
Total Cyanide	<0.005 ND	NA	---	NA
Alkalinity (Total)	130	2081.66	---	NA
Hardness (Calculated)	210	3362.69	---	NA
Calcium	58	928.74	---	NA
Potassium	14	224.18	---	NA
Magnesium	17	272.22	---	NA
Chloride	150	2401.92	---	NA
Sulfate	57	912.73	---	NA
	µg/L	lbs/day	µg/L	lbs/day
Total Arsenic (As)	<1 ND	NA	---	NA
Total Cadmium (Cd)	<0.2 ND	NA	<0.2 *	NA
Total Chromium (Cr)	<1 ND	NA	---	NA
Total Copper (Cu)	6.8	0.11	6.9 *	0.11
Total Lead (Pb)	<1 ND	NA	<1.0 *	NA
Total Mercury (Hg)	<0.2 ND	NA	<0.00025*^	NA
Total Selenium (Se)	<1 ND	NA	<5.0 *	NA
Total Silver (Ag)	<0.2 ND	NA	---	NA
Total Zinc (Zn)	25	0.40	12.7 *	0.20
Total Nickel (Ni)	3.2	0.05	<5.0 *	NA
Total Molybdenum (Mo)	<5 ND	NA	---	NA

NOTE:

* - Facility results from 7/6/2023

^ - units converted from ng/L to µg/L

Table 2 - Grab Samples

Monitoring Point	EGLE				Cadillac WWTP	
	001	001	001	Upstream	001	001
	Date Collected	7/11/2023	7/12/2023	7/11/2023	7/11/2023	7/12/2023
Time Collected	10:15	14:34	9:34	13:57	7:00	7:00
Results				Results		
Temperature (C)	19.5	20.5	19.5	18.4	---	---
pH (S.U.)	7.82	7.83	7.95	8	8.03	8.01
Dissolved Oxygen (mg/L)	8.35	8.45	8.51	8.86	8.02	7.99
Fecal Coliform (MPN/100 ml)	---	---	<10 ND	---	2	2
mg/L				mg/L		
Alkalinity (Total)	---	---	---	130	---	---
Hardness (Calculated)	---	---	---	160	---	---
Calcium	---	---	---	46	---	---
Magnesium	---	---	---	10	---	---
Chloride	---	---	---	60	---	---
Sulfate	---	---	---	12	---	---
ng/L				ng/L		
PFOS	---	---	<2 ND	---	---	---
PFOA	---	---	2.9	---	---	---
PFBS	---	---	22	---	---	---
PFNA	---	---	<2 ND	---	---	---
PFHxS	---	---	<2 ND	---	---	---
PFHxA	---	---	12	---	---	---
HFPO-DA	---	---	<4 ND	---	---	---

Table 3 - BNA

Monitoring Point	EGLE 001	Facility Composite 001	Facility 001
Sample Method	Grab Comp	Comp *	---
Date Collected	7/12/2023	7/12/2023	7/6/2023
Time Collected	9:45	7:00	---
BNA Scan Semivolatiles (Method 625)	µg/L	µg/L	µg/L
1,2,4-Trichlorobenzene	<2.0 ND	<2.0 ND	---
2,4-Dinitrotoluene	<5.0 ND	<5.0 ND	---
2,6-Dinitrotoluene	<5.0 ND	<5.0 ND	---
2-Chloroaniline	<5.0 ND, A08	<5.0 ND, A08	---
2-Chloronaphthalene	<2.0 ND	<2.0 ND	---
2-Methylnaphthalene	<5.0 ND, A07	<5.0 ND, A07	---
2-Nitroaniline	<20 ND	<20 ND	---
3-Nitroaniline	<20 ND	<20 ND	---
4-Bromophenyl phenyl ether	<2.0 ND	<2.0 ND	---
4-Chloroaniline	<10 ND	<10 ND	---
4-Chlorodiphenylether	<1.0 ND	<1.0 ND	---
4-Nitroaniline	<20 ND	<20 ND	---
Acenaphthene	<1.0 ND	<1.0 ND	---
Acenaphthylene	<1.0 ND	<1.0 ND	---
Aniline	<4.0 ND	<4.0 ND	---
Anthracene	<1.0 ND	<1.0 ND	---
Azobenzene	<2.0 ND	<2.0 ND	---
Benzo(a)anthracene	<1.0 ND	<1.0 ND	---
Benzo(a)pyrene	<1.0 ND	<1.0 ND	---
Benzo(b)fluoranthene	<1.0 ND	<1.0 ND	---
Benzo(g,h,i)perylene	<1.0 ND	<1.0 ND	---
Benzo(k)fluoranthene	<1.0 ND	<1.0 ND	---
Benzyl Alcohol	<50 ND	<50 ND	---
Bis(2-chloroethoxy)methane	<2.0 ND	<2.0 ND	---
Bis(2-chloroethyl)ether	<1.0 ND	<1.0 ND	---
Bis(2-chloroisopropyl)ether	<1.0 ND	<1.0 ND	---
Bis(2-ethylhexyl)phthalate	<5.0 ND	<5.0 ND	<3.9
Butyl benzyl phthalate	<5.0 ND	<5.0 ND	---
Carbazole	<5.0 ND	<5.0 ND	---
Chrysene	<1.0 ND	<1.0 ND	---
Dibenz(a,h)anthracene	<2.0 ND	<2.0 ND	---
Dibenzofuran	<4.0 ND	<4.0 ND	---
Diethylphthalate	<5.0 ND	<5.0 ND	---
Dimethyl phthalate	<5.0 ND	<5.0 ND	---
Di-n-butyl phthalate	<5.0 ND	<5.0 ND	---
Di-n-octyl phthalate	<5.0 ND	<5.0 ND	---
Fluoranthene	<1.0 ND	<1.0 ND	---
Fluorene	<1.0 ND	<1.0 ND	---
Hexachlorobenzene	<2.0 ND	<2.0 ND	---

Table 3 - BNA

	EGLE	Facility Composite	Facility
Monitoring Point	001	001	001
Sample Method	Grab Comp	Comp *	---
Date Collected	7/12/2023	7/12/2023	7/6/2023
Time Collected	9:45	7:00	---
BNA Scan Semivolatiles, continued	µg/L	µg/L	µg/L
Hexachlorobutadiene	<2.0 ND	<2.0 ND	---
Hexachlorocyclopentadiene	<10 ND, A05, A08	<10 ND, A05, A08	---
Hexachloroethane	<1.0 ND	<1.0 ND	---
Indeno(1,2,3-c,d)pyrene	<2.0 ND	<2.0 ND	---
Isophorone	<1.0 ND	<1.0 ND	---
N, N-dimethylaniline	<5.0 ND	<5.0 ND	---
Napthalene	<1.0 ND, A07	<1.0 ND, A07	---
Nitrobenzene	<2.0 ND	<2.0 ND	---
N-methylaniline	<1.0 ND	<1.0 ND	---
N-Nitrosodimethylamine	<5.0 ND, A05	<5.0 ND, A05	---
N-Nitrosodi-n-propylamine	<2.0 ND	<2.0 ND	---
N-Nitrosodiphenylamine	<2.0 ND	<2.0 ND	---
Phenanthrene	<1.0 ND	<1.0 ND	---
Pyrene	<1.0 ND	<1.0 ND	---
Tetramethylurea	<1.0 ND	<1.0 ND	---

NOTES:

* Facility Composite BNA was collected by facility but run at the EGLE lab as part of the facility composite collected by EGLE staff.

Table 4 - 001 Split

	Composite Split		EGLE Composite Split	
Monitoring Point	001		001	
Date Collected	7/12/2023		7/12/2023	
Time Collected	7:00		9:45	
	EGLE Results	Facility Results	EGLE Results	Facility Results
	mg/L	mg/L	mg/L	mg/L
Suspended Solids	<4 ND	4.4	4	6.4
CBOD-5	<3 ND	3	<3 ND	<3
BOD-5	3.8	---	<3 ND	---
COD	37	---	37	---
TOC	13	---		---
NO2	0.078	---	0.012	---
NO3 + NO2	8	---	9.3	---
NH3	0.02	<0.30		<0.3
Kjeldahl Nitrogen	1.3	---	1.2	---
Total Phosphorus	0.39	0.3		0.32
Ortho Phosphate	0.21	---	0.23	---
Total Cyanide	<0.005 ND	---	<0.005 ND	---
Alkalinity (Total)	120	---	130	---
Hardness (Calculated)	210	---	210	---
Calcium	58	---	58	---
Potassium	13	---	14	---
Magnesium	17	---	17	---
Chloride	150	---	150	---
Sulfate	57	---	57	---
	µg/L	µg/L	µg/L	µg/L
Total Arsenic (As)	<1 ND	---	<1ND	---
Total Cadmium (Cd)	<0.2 ND	<0.2*	<0.2 ND	<0.2
Total Chromium (Cr)	<1 ND	---	<1 ND	---
Total Copper (Cu)	6.9	6.9*	6.8	6.7
Total Lead (Pb)	<1 ND	<1.0*	<1 ND	<1
Total Mercury (Hg)	<0.2 ND	<0.00025*^	<0.2 ND	<0.091
Total Selenium (Se)	<1 ND	<5.0*	<1 ND	<5
Total Silver (Ag)	<0.2 ND	---	<0.2 ND	---
Total Zinc (Zn)	18	12.7*	25	22.7
Total Nickel (Ni)	2.4	<5.0*	3.2	<5
Total Molybdenum (Mo)	<5 ND	---	<5 ND	---

NOTES:

*-Facility results from 7/6/23

^ - Sample converted from ng/L to µg/L

Table 5 - Comparison of Results with Previous Survey

Monitoring Point	001	
Survey Period Start Date/Time	5/9/18 10:20	7/11/23 10:00
Survey Period End Date/Time	5/10/18 9:20	7/12/23 9:45
	EGLE Results	
	mg/L	mg/L
Suspended Solids	7	4
CBOD-5	<3 ND	<3 ND
BOD-5	---	<3 ND
COD	---	37
TOC	---	12
NO2	---	0.012
NO3 + NO2	---	9.3
NH3	0.12	0.03
Total Phosphorus	0.15	0.4
Ortho Phosphate	---	0.23
Total Cyanide	---	<0.005 ND
Available Cyanide	<0.002 ND	---
Alkalinity (Total)	130	130
Hardness (Calculated)	---	210
Calcium	---	58
Potassium	---	14
Magnesium	---	17
Chloride	---	150
Sulfate	---	57
	µg/L	µg/L
Total Arsenic (As)	<1 ND	<1 ND
Total Barium (Ba)	23	---
Total Cadmium (Cd)	<0.2 ND	<0.2 ND
Total Chromium (Cr)	1.2	<1 ND
Total Copper (Cu)	6.4	6.8
Total Lead (Pb)	<1 ND	<1 ND
Total Mercury (Hg)	<0.2 ND	<0.2 ND
Total Selenium (Se)	<1 ND	<1 ND
Total Silver (Ag)	<0.2 ND	<0.2 ND
Total Zinc (Zn)	20	25
Total Nickel (Ni)	4.5	3.2
Total Molybdenum (Mo)	12	<5 ND

Table 6 - Sample Preservation and Lab Letter Codes

Parameter	Preservative
COD/TOC/Phenol/Nutrients (Chlorine Absent)	10 drops H2SO4/500 ml (to pH <2)
Cyanide/Thiocyanates	10 drops NaOH (to pH >12)/250 ml
pH & Dissolved Oxygen (D.O.)	Meter reading
Total Metals	5ml HNO3/500 ml (to pH <2)
Microbiology	prepreserved bottle with sodium thiosulfate
Samples preserved as required, cooled to 6 degrees Centigrade with chain of custody maintained	

Lab Letter Codes

ND - Indicates compound analyzed for but not detected

A05 - Result(s) and reporting limit(s) are estimated due to low continuing calibration standard criteria failure.

A07 - Result(s) and reporting limit(s) are estimated due to poor precision.

A08 - Result(s) and reporting limit(s) are estimated due to low recovery of batch QC.

Survey by: Ashley McElmurry, Environmental Quality Analyst

Contact with Management: Cindy Tomaszewski and Patrick Miller

Laboratory Analyses by: Michigan Department of Environment, Great Lakes and
Energy, Lansing, Michigan
Merit Laboratories, Inc. of East Lansing, Michigan

Report by: Ashley McElmurry, Environmental Quality Analyst
Field Operations Section
Water Resources Division

Publish Date: 22-Aug-23